

PERINPANAYAGAM SIVASANKAR

Software Engineer (Trainee) | Data Science Undergraduate

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PROFESSIONAL SUMMARY

Data Science undergraduate at Sri Lanka Institute of Information Technology with hands-on experience in data analytics, AI-driven systems, full-stack web development, and IoT solutions using Python, React, MongoDB, and scikit-learn. Delivered 7 end-to-end academic projects including predictive analytics models, NLP-based AI systems, and web applications. Skilled in machine learning, data visualization, and backend development, with a strong interest in Data Science, AI, and software engineering internships.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++

Frontend: React.js, HTML5, CSS3

Backend: Express.js, Node.js, REST APIs, Flask

Databases: MySQL, MongoDB, SQL

Tools & Platforms: Git, GitHub, Arduino IDE, VS Code

Core Concepts: Data Structures, Algorithms, Machine Learning, Data Analysis, OOP, Agile

PROJECTS

Arduino UNO Smart Greenhouse System github.com/Sankarboy200313 Jun – Dec 2024

- Designed an IoT-based greenhouse monitoring system using Arduino UNO, DHT11, and soil moisture sensors
- Implemented automated relay-based irrigation when moisture levels dropped below a configurable threshold
- Logged real-time temperature, humidity, and soil moisture data at 1Hz for environmental analysis

DCN Network Architecture Proposal github.com/Sankarboy200313 Jun – Dec 2024

- Designed a secure multi-tier campus network architecture for a 3-building environment using Cisco Packet Tracer
- Configured VLANs, CIDR subnetting, and inter-VLAN routing to segment 4 departments
- Documented security controls including ACLs and DMZ zones to minimise lateral movement risks

Online Bakery Product Platform github.com/Sankarboy200313 Feb – Jun 2025

- Built a full-stack e-commerce platform using React.js, Express.js, and MongoDB
- Implemented product CRUD operations, cart management, and order tracking for 30+ products
- Optimised MongoDB queries with indexing, reducing average product loading time by approximately 40%

Hospital Management System github.com/Sankarboy200313 Jul – Dec 2025

- Engineered a full-stack hospital management system using React.js and Node.js/Express REST APIs
- Designed a MySQL relational database schema to manage patient records, appointments, and staff workflows
- Implemented role-based access control for admins, doctors, and receptionists to improve data security
- Developed a responsive user interface using CSS Grid and Flexbox

AI Prediction Model github.com/Sankarboy200313 Jul – Dec 2025

- Developed a supervised machine learning classification model using Python and scikit-learn (Random Forest)
- Applied preprocessing techniques including null-value imputation, label encoding, and feature scaling
- Improved model accuracy by 12% over baseline performance
- Evaluated performance using confusion matrix, precision, recall, and F1-score metrics

Mental Wellbeing AI System github.com/Sankarboy200313 Feb – Jun 2026

- Built an NLP-powered wellbeing recommendation system using Python and scikit-learn

- Developed a rule-based intent classifier achieving approximately 83% accuracy on a 200-sample dataset
- Designed a Flask REST API delivering personalised wellbeing recommendations with sub-200ms response latency
- Integrated MongoDB to maintain adaptive mood-based user profiles across sessions

Hospital Management Mobile Application github.com/Sankarboy200313

Feb – Jun 2026

- Developed a cross-platform mobile application using React Native and MongoDB Atlas
- Implemented offline-first patient record synchronisation with server reconciliation upon reconnection
- Reduced appointment booking workflow steps from 6 to 3 through UI redesign improvements
- Integrated RESTful APIs with JWT authentication for secure access control

EXPERIENCE

Academic Project Developer | SLIIT — Sri Lanka Institute of Information Technology 2024 – Present

- Delivered 7 academic software projects applying Python, React, SQL, MongoDB, and Arduino in practical implementations
- Collaborated in 3-to-5 person agile teams, participating in sprint planning, code reviews, and demo presentations
- Strengthened system design skills by independently architecting database schemas and REST API structures for each project

EDUCATION

BSc (Hons) in Data Science | Sri Lanka Institute of Information Technology (SLIIT) 2024 – 2028 (Expected)

- Relevant coursework: Data Structures & Algorithms, Database Systems, Web Development, Object-Oriented Programming, Networking

GCE Advanced Level (A/L) | Jaffna Hindu College

2022

CERTIFICATIONS

- LinkedIn Learning – AI Basics
- University of Moratuwa – Python for Beginners
- Gnanam Education – Programming Basics
- CB Education – Programming & English
- Forage - Tata Data Visualisation: Empowering Business with Effective Insights
- Forage - Tata GenAI Powered Data Analytics

LANGUAGES

- Tamil (Native), English (Professional), Sinhala (Conversational)